

BAYES AND ROBUST BAYES PREDICTION WITH AN APPLICATION TO AIR TEMPERATURE PREDICTION

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Summary: Prediction of a future observation on the basis of currently observed data is demanded in many theoretical and applied problems. In this paper, we introduce prediction of a future observation from scale models under the general entropy prediction loss function and deal with Bayes and Posterior Regret Gamma Minimax prediction and obtain general form of the predictors. We study and discuss the air temperature prediction problem and utilize the results using an available well-known data set. A simulation study is also provided to illustrate the practical utility of the prediction procedure.

1. Introduction

Prediction of a future observation based on currently observed data arises in many principle areas of statistical research and have been very useful in many fields of application such as quality control, life experiments and so on. Prediction problems considered by both frequentists and Bayesians are abundant in the literature, see for examples Eaton (1992), Bjornstad (1996), Nayak (2000), Wang and Veraverbeke (2009) and references therein.

Let X be an observed random variable (real or vector-valued) and Y be a future real random variable with the joint density function $f(\cdot|\eta)$. Further suppose η has a prior distribution Π and density π w.r.t. some σ -finite measure ν . The goal is to predict some future observation of Y on the basis of currently observed x . The predictive distribution is given by

$$h_{\pi}(y|x) = \int_{\Theta} f(y|x, \eta) \pi(\eta|x) d\nu(\eta) = \int_{\Theta} \frac{1}{f^{*}(x|\eta)} f(x, y|\eta) \pi(\eta|x) d\nu(\eta), \quad (1)$$

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where $f^*(x|\eta)$ is the marginal density of X given η and $\pi(\eta|x)$ is the posterior density function of η given $X = x$. In some cases, it is reasonable to assume that given η , X and Y are conditionally independent, i.e., $f(y|x, \eta) = f(y|\eta)$, see Spiegelhalter, Abrams and Myles (2004) and Boratynska (2006). In this case, the predictive distribution (1) reduces to

$$h_\pi(y|x) = \int_{\Theta} f(y|\eta) \pi(\eta|x) d\nu(\eta).$$

Such predictive distributions are useful in many contexts: Grieve (1988) provides examples in bio-equivalence, trial monitoring and toxicology while Berry and Stangl (1996) describe their use in design and power calculations, model checking, and in deciding whether to conduct a future trial. For examples of explicit predictive probability density function (pdf) for regression and other models, see Guttman (1970), Zellner (1971), Box and Tiao (1973) and Aitchison and Dunsmore (1975). Applications of prediction including power calculations, sequential analysis, health policy-making, and payback from research are given in Spiegelhalter et al. (2004).

Prediction of a future observation has been recently developed under different prediction loss functions. Jafari Jozani and Parsian (2008) considered posterior regret Gamma-minimax prediction based on k -record data under Entropy Prediction Loss Function (EPLF)

$$L(y, d) = \frac{y}{d} - \ln\left(\frac{y}{d}\right) - 1. \quad (2)$$

This loss was first considered by Dey, Ghosh and Srinivasan (1987) and Dey and Liu (1992). Kiapour and Nematollahi (2011) discussed the problem of robust Bayesian prediction under squared log error prediction loss function

$$L(y, d) = (\ln y - \ln d)^2 = \left(\ln \frac{y}{d}\right)^2.$$

This loss was proposed by Brown (1968) and known as balanced loss. See also Ahmadi, Doostparast and Parsian (2005) and Boratynska (2006) for prediction under LINEX prediction loss, Boratynska (2005) for prediction under quadratic prediction loss function and Ahmadi, MirMostafaei and Balakrishnan (2012) for prediction under squared error, zero-one and absolute error prediction loss functions.

In this paper, we consider the problem of obtaining some optimal predictors of Y under the General Entropy Prediction Loss Function (GEPLF)

$$L(y, d) = \left(\frac{y}{d}\right)^q - q \ln \frac{y}{d} - 1, \quad q \neq 0, \quad (3)$$

which is asymmetric and convex in $\Delta = \frac{y}{d}$ when $q = 1$ and quasi-convex otherwise, has a unique minimum at $\Delta = 1$, is scale invariant and also is useful in situations where underprediction and overprediction have not a same penalty. The graphs of GEPLF for $q = -2, -1, 1, 2$ are depicted in Figure 1. The GEPLF with negative q values penalizes overprediction more than underprediction while with positive q values acts vice-versa.

It is worth mentioning that near $\Delta = 1$

$$\left(\frac{y}{d}\right)^q - q \ln \frac{y}{d} - 1 \simeq \frac{1}{2} q^2 \left(\frac{y}{d} - 1\right)^2$$

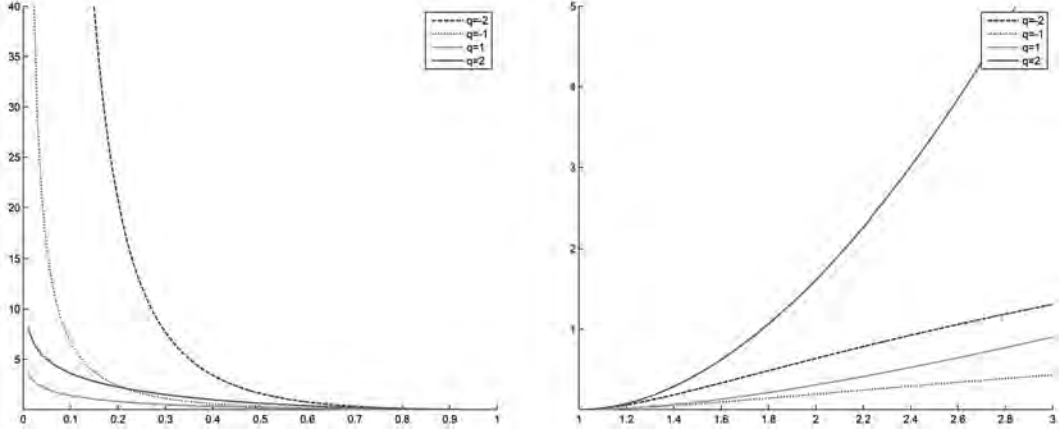


Figure 1: The GEPLF with different q values.

and for small $|q|$ values,

$$\left(\frac{y}{d}\right)^q - q \ln \frac{y}{d} - 1 \simeq \frac{1}{2} q^2 (\ln y - \ln d)^2.$$

It is worth mentioning that the GEPLF was first considered by Calabria and Pulcini (1994) and applied by others thereafter. However, all of the works in the literature have been done for estimation purposes and to the best of our knowledge, this paper is the first work in prediction area considering the GEPLF.

To have range-preserved predictors and due to the structure of GEPLF, we consider predictors d of Y based on X to satisfy

$$P_\eta(d(X) \in S_Y^*) = 1, \quad \forall \eta \in \Theta,$$

where S_Y^* is the support of Y . This class of predictors is denoted by $\mathcal{D}$.

The structure of this paper is as follows. In Sect. 2, we discuss Bayes and Posterior Regret Gamma Minimax predictions w.r.t. ε -contaminated priors. An air temperature prediction problem using a real data set and a simulation study are provided in Sect. 3. Finally, we end with concluding remarks in Sect. 4.

2. Bayes and Robust Bayes Prediction

In this section, we discuss the prediction problem using Bayesian view point which has an important rule in modern statistics (see Berger, 1985) and obtain the Bayes and robust Bayes predictions of Y under the GEPLF. Let $r(\pi, d)$ be the Bayes risk of the predictor d under the prior π . We say $d_\pi(X)$ is a Bayes predictor of Y if $r(\pi, d_\pi) \leq r(\pi, d)$, for all $d \in \mathcal{D}$, where

$$r(\pi, d) = E \left[\left(\frac{Y}{d} \right)^q - q \ln \frac{Y}{d} - 1 \right] = \int \rho(\pi, d) f_0(x) dx,$$

$f_0(x)$ is the marginal density of X and

$$\begin{aligned}\rho(\pi, d) &= \int \int \left(\left(\frac{y}{d} \right)^q - q \ln \frac{y}{d} - 1 \right) f(y|x, \eta) \pi(\eta|x) dy d\nu(\eta) \\ &= E \left[\left(\frac{Y}{d} \right)^q - q \ln \frac{Y}{d} - 1 | X = x \right]\end{aligned}\quad (4)$$

is the posterior prediction risk over $d \in \mathcal{D}$. Obviously for a given x , minimizing $\rho(\pi, d)$ leads us to minimizing $r(\pi, d)$. Hence, if $X = x$, then the Bayes predictor is obtained by minimizing the posterior prediction risk over $d \in \mathcal{D}$. The following proposition is useful in our further developments.

Proposition 1 The unique Bayes predictor of Y w.r.t. some prior π , say $d_\pi(X)$, under the GEPLF is given by

$$d_\pi(X) = E^{\frac{1}{q}} [Y^q | X = x]. \quad (5)$$

Any Bayesian prediction problem is treated by assigning a specified prior distribution π over the parameter space while in practice, the specified prior knowledge is rare and no one can say a chosen prior is better than any other. Robust Bayes analysis which deals with the problem of explaining uncertainty of the inputs (the prior, the model, and the loss function), provides a breakthrough way to take into account the input's variation. If the uncertainty is in terms of the prior knowledge, robust Bayes analysis provides a way to consider the prior knowledge in terms of a class of priors Γ and compute a posterior functional, such as posterior risk, Bayes risk or posterior expected value, as the prior ranges over Γ . In fact robust rules are aimed at global prevention against inappropriate choices of priors or its hyperparameters. See Berger (1984), Berger (1985), and Berger (1990) for more details.

In the context of robust Bayes prediction dealing with the posterior regret due to choosing the predictor d instead of the Bayes predictor d_π , i.e., $r_p(d, d_\pi) = \rho(\pi, d) - \rho(\pi, d_\pi)$, the predictor $d_{PRGM, \Gamma}$ is said to be Posterior Regret Gamma Minimax (PRGM) predictor if

$$r_p(d_{PRGM, \Gamma}, d_\pi) = \inf_{d \in \mathcal{D}} \sup_{\pi \in \Gamma} r_p(d, d_\pi).$$

Regret-type rules in decision theory has been used and appreciated for a very long time. See Berger (1984) and Berger (1985) for an excellent discussion on the topic. The context of conditional Gamma minimax regret rules was introduced by Zen and DasGupta (1993).

In connection with prior uncertainty, it is interesting to consider an ε -contaminated class of priors given by

$$\Gamma_\varepsilon = \{ \pi_\varepsilon = (1 - \varepsilon)\pi_0 + \varepsilon\pi : \pi \in \Gamma, \varepsilon \in [0, 1] \}, \quad (6)$$

where π_0 is an explicit prior and $\pi \in \Gamma$, is any prior in the family of all probability distributions on $(0, +\infty)$ and $\varepsilon \in [0, 1]$ is a fixed known number. The ε -contaminated class of priors Γ_ε is a beneficial tool when dealing with the effect of prior uncertainty, see Berger (1985), Sivaganesan (1988) and Sivaganesan and Berger (1989) among others.

Assuming $\pi_\varepsilon \in \Gamma_\varepsilon$ as a prior given in (6), the posterior risk of a predictor $d(X)$ under GEPLF (3) is given by

$$\rho(\pi_\varepsilon, d) = E[L(Y, d)|X] = \frac{E[Y^q|X]}{d^q} + q \ln d - qE[\ln Y|X] - 1,$$

which leads to the following posterior regret function

$$\begin{aligned} r_p(d, d_{\pi_\varepsilon}) &= \rho(\pi_\varepsilon, d) - \rho(\pi_\varepsilon, d_{\pi_\varepsilon}) \\ &= \frac{d_{\pi_\varepsilon}^q}{d^q} - q \ln \frac{d_{\pi_\varepsilon}}{d} - 1. \end{aligned} \quad (7)$$

In the following theorem, we obtain PRGM predictor of Y under the GEPLF (3).

Theorem 1 Let $\underline{d} = \underline{d}(X) = \inf_{\pi_\varepsilon \in \Gamma_\varepsilon} d_{\pi_\varepsilon}(X)$ and $\bar{d} = \bar{d}(X) = \sup_{\pi_\varepsilon \in \Gamma_\varepsilon} d_{\pi_\varepsilon}(X)$ be finite and $\underline{d} < \bar{d}$. Then the PRGM predictor of Y under the GEPLF (3) is

$$d_{PRGM, \Gamma_\varepsilon}(X) = \left(\frac{\bar{d}^q(X) - \underline{d}^q(X)}{\ln \bar{d}^q(X) - \ln \underline{d}^q(X)} \right)^{1/q}.$$

Proof. See the Appendix.

The following lemma plays a key role in applying the results in our next examples.

Lemma 1 Let X denote a data set with pdf $f(\cdot|\eta)$, $\eta \in \Theta$ and consider some mixed prior $\pi_\varepsilon(\eta) = (1 - \varepsilon)\pi_0(\eta) + \varepsilon\pi(\eta)$. Then,

(i) The posterior distribution is $\pi_\varepsilon(\eta|x) = \lambda(x)\pi_0(\eta|x) + [1 - \lambda(x)]\pi(\eta|x)$ where

$$\lambda(x) = \frac{(1 - \varepsilon)m_0(x)}{(1 - \varepsilon)m_0(x) + \varepsilon m^*(x)},$$

$m_0(x) = \int_\Theta f(x|\eta)\pi_0(\eta)d\nu(\eta)$, $m^*(x) = \int_\Theta f(x|\eta)\pi(\eta)d\nu(\eta)$ and $\nu(\cdot)$ is a σ -finite measure,

(ii) The predictive density (1) is $h_{\pi_\varepsilon}(y|x) = \lambda(x)h_{\pi_0}(y|x) + [1 - \lambda(x)]h_\pi(y|x)$,

(iii) The Bayes predictor is $d_{\pi_\varepsilon}(X) = \sqrt[q]{\lambda(X)d_{\pi_0}(X) + (1 - \lambda(X))d_\pi(X)}$.

The following examples clarify how to deal with Bayes and robust Bayes prediction in Gamma and Exponential models, respectively.

Example 1 Let $X_1, X_2, \dots$ be a sequence of independent random variables from $\Gamma(\alpha, \eta)$ -distribution with pdf

$$f(x|\eta) = \frac{1}{\Gamma(\alpha)} \eta^{-\alpha} x^{\alpha-1} e^{-\frac{x}{\eta}}, \quad x > 0, \alpha > 0, \eta > 0,$$

where the shape parameter α is known and the scale parameter η is unknown. Based on observing $X_1, \dots, X_n$ with summarized information in $X = \sum_{i=1}^n X_i \sim \Gamma(\alpha_1, \eta)$, where $\alpha_1 = n\alpha$, we are interested in predicting $Y = \sum_{i=n+1}^m X_i \sim \Gamma(\alpha_2, \eta)$. Let $\pi_\varepsilon(\eta) = (1 - \varepsilon)\pi_0(\eta) + \varepsilon\pi(\eta)$, where π_0 and $\pi \in \Gamma$ having $\Pi(\gamma_0, \frac{1}{g_0})$ and $\Pi(\gamma, \frac{1}{g})$ distributions respectively, and Γ be one of the following three classes:

$$\begin{aligned} \Gamma_\gamma &= \left\{ \Pi(\gamma, \frac{1}{g}); g = g_0, \gamma \in [\gamma_1, \gamma_2] \subseteq \mathfrak{R}^+ \right\}, \text{ where } \gamma_1 < \gamma_2 \text{ and } g_0 \text{ are fixed;} \\ \Gamma_g &= \left\{ \Pi(\gamma, \frac{1}{g}); \gamma = \gamma_0, g \in [g_1, g_2] \subseteq \mathfrak{R}^+ \right\}, \text{ where } g_1 < g_2 \text{ and } \gamma_0 \text{ are fixed;} \end{aligned}$$

$\Gamma_{\gamma,g} = \left\{ \Pi(\gamma, \frac{1}{g}); \gamma \in [\gamma_1, \gamma_2] \subseteq \Re^+, g \in [g_1, g_2] \subseteq \Re^+ \right\}$, where $\gamma_1 < \gamma_2$ and $g_1 < g_2$ are fixed.

Then,

$$\pi_\varepsilon(\eta|x) = \lambda(x) \Pi(\alpha_1 + \gamma_0, \frac{1}{x+g_0}) + (1 - \lambda(x)) \Pi(\alpha_1 + \gamma, \frac{1}{x+g})$$

with $\lambda(x) = \frac{1-\varepsilon}{1-\varepsilon+\varepsilon m^*(x)}$ and $m^*(x) = \frac{\Gamma(\gamma_0)\Gamma(\alpha_1+\gamma)}{\Gamma(\gamma)\Gamma(\alpha_1+\gamma_0)} \left(\frac{g_0+x}{g+x} \right)^{\alpha_1-1} \left(\frac{g}{g+x} \right)^\gamma \left(\frac{g_0}{g_0+x} \right)^{-\gamma_0}$. The Bayes predictor w.r.t. π_ε is given by

$$d_{\pi_\varepsilon}(X) = d_q^*(\gamma, g) = \sqrt[q]{\lambda(X) d_q(\gamma_0, g_0) + (1 - \lambda(X)) d_q(\gamma, g)},$$

where $d_q(\gamma, g) = \psi^q(\alpha_1 + \gamma, \alpha_2, q)(X + g)^q$, $\psi(\alpha_1, \alpha_2, q) = \sqrt[q]{\frac{\Gamma(\alpha_1 - q)\Gamma(\alpha_2 + q)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}}$, $\alpha_1 - q > 0$ and $\alpha_2 + q > 0$. Hence, when $\varepsilon = 0$, the Bayes predictor of Y is $d_q^*(\gamma, g) = \psi(\alpha_1 + \gamma_0, \alpha_2, q)(X + g_0)$ and when $\varepsilon = 1$, it is $d_q^*(\gamma, g) = \psi(\alpha_1 + \gamma, \alpha_2, q)(X + g)$. The PRGM predictors of Y under the GEPLF (3) for the three class of priors in Γ_g , Γ_γ and $\Gamma_{\gamma,g}$ are summarized below

$$d_{PRGM, \Gamma} = \begin{cases} \left(\frac{d_q^{*q}(\gamma_0, g_2) - d_q^{*q}(\gamma_0, g_1)}{\ln d_q^{*q}(\gamma_0, g_2) - \ln d_q^{*q}(\gamma_0, g_1)} \right)^{\frac{1}{q}} & \Gamma = \Gamma_g, \\ \left(\frac{d_q^{*q}(\gamma_1, g_0) - d_q^{*q}(\gamma_2, g_0)}{q(\ln d_q^{*q}(\gamma_1, g_0) - \ln d_q^{*q}(\gamma_2, g_0))} \right)^{\frac{1}{q}} & \Gamma = \Gamma_\gamma, \\ \left(\frac{d_q^{*q}(\gamma_1, g_2) - d_q^{*q}(\gamma_2, g_1)}{q(\ln d_q^{*q}(\gamma_1, g_2) - \ln d_q^{*q}(\gamma_2, g_1))} \right)^{\frac{1}{q}} & \Gamma = \Gamma_{\gamma,g}. \end{cases}$$

Example 2 Let $X_1, X_2, \dots$ be a sequence of *i.i.d* random variables from an Exponential distribution with pdf

$$f(x|\eta) = \frac{1}{\eta} e^{-x/\eta}, \quad x > 0, \eta > 0.$$

Following Arnold, Balakrishnan and Nagaraja (1998), an upper k -record process is in fact k -th largest X yet seen. For a formal definition, consider $X_{i:m}$ denotes the i th order statistics in a sample of size m and let $T_{1(k)} = k$, $R_{1(k)} = X_{1:k}$ and for $n \geq 2$

$$T_{n,k} = \min \left\{ j : j > T_{n-1,k}, X_j > X_{T_{n,k}-k+1:T_{n-1,k}} \right\},$$

then, $R_{n(k)} = X_{T_{n,k}-k+1:T_{n-1,k}}$, $n \geq 1$ is the upper k -record. Also, the joint density of $(r_{n(k)}, r_{m(k)})$, $m > n$, given η is

$$f(r_{n(k)}, r_{m(k)}|\eta) = \frac{(r_{m(k)} - r_{n(k)})^{m-n-1}}{\Gamma(n)\Gamma(m-n)} \left(\frac{k}{\eta} \right)^m r_{n(k)}^{n-1} e^{-\frac{k}{\eta} r_{m(k)}}, \quad r_{m(k)} > r_{n(k)}. \quad (8)$$

Also, the marginal pdf of $R_{n(k)}$ is

$$f(r_{n(k)}|\eta) = \frac{1}{\Gamma(n)} \left(\frac{k}{\eta} \right)^n r_{n(k)}^{n-1} e^{-\frac{k}{\eta} r_{n(k)}}, \quad r_{n(k)} > 0, \eta > 0. \quad (9)$$

Assuming $\pi_\varepsilon(\eta) = (1 - \varepsilon)\pi_0(\eta) + \varepsilon\pi(\eta)$ -prior, where π_0 and $\pi \in \Gamma$ having $\Pi(\gamma_0, \frac{1}{g_0})$ and $\Pi(\gamma, \frac{1}{g})$ distributions respectively and Γ is one of three classes in Γ_g , Γ_γ and $\Gamma_{\gamma,g}$. It is easy to verify that the

posterior distribution of η given $r_{n(k)}$ is

$$\pi_{\varepsilon}(\eta|r_{n(k)}) = \lambda(r_{n(k)}) \Pi(n + \gamma_0, \frac{1}{g_0 + kr_{n(k)}}) + (1 - \lambda(r_{n(k)})) \Pi(n + \gamma, \frac{1}{g + kr_{n(k)}}),$$

with $\lambda(r_{n(k)}) = \frac{1-\varepsilon}{1-\varepsilon+\varepsilon m^*(r_{n(k)})}$ and

$$m^*(r_{n(k)}) = \frac{\Gamma(\gamma_0)\Gamma(n+\gamma)}{\Gamma(\gamma)\Gamma(n+\gamma_0)} \left(\frac{g_0 + kr_{n(k)}}{g + kr_{n(k)}} \right)^{n-1} \left(\frac{g}{g + kr_{n(k)}} \right)^{\gamma} \left(\frac{g_0}{g_0 + kr_{n(k)}} \right)^{-\gamma_0}.$$

Then, the predictive density w.r.t. the prior π_{ε} reduces to

$$\begin{aligned} f(r_{m(k)}|r_{n(k)}) &= \lambda(r_{n(k)}) \frac{k^{m-n}\Gamma(m+\gamma_0)}{\Gamma(m-n)\Gamma(n+\gamma_0)} \frac{(g_0 + kr_{n(k)})^{n+\gamma_0}}{(g_0 + kr_{m(k)})^{n+\gamma_0}} \frac{(r_{m(k)} - r_{n(k)})^{m-n-1}}{(g_0 + kr_{m(k)})^{m-n}}, \\ &+ (1 - \lambda(r_{n(k)})) \frac{k^{m-n}\Gamma(m+\gamma)}{\Gamma(m-n)\Gamma(n+\gamma)} \frac{(g + kr_{n(k)})^{n+\gamma}}{(g + kr_{m(k)})^{n+\gamma}} \frac{(r_{m(k)} - r_{n(k)})^{m-n-1}}{(g + kr_{m(k)})^{m-n}}, \end{aligned}$$

provided $r_{m(k)} > r_{n(k)}$. After some algebraic manipulations, we observe that given $R_{n(k)} = r_{n(k)}$, $Z_0 = \frac{g_0 + kR_{n(k)}}{g_0 + kR_{m(k)}} \sim \text{Beta}(n + \gamma_0, m - n)$, $Z = \frac{g + kR_{n(k)}}{g + kR_{m(k)}} \sim \text{Beta}(n + \gamma, m - n)$ and the Bayes predictor of $R_{m(k)}$ under the GEPLF (3) is given by

$$BR_{m(k)} = \Psi_q^*(\gamma, g) = \sqrt[q]{\lambda(r_{n(k)})\Psi_q(\gamma_0, g_0) + (1 - \lambda(r_{n(k)}))\Psi_q(\gamma, g)},$$

where $\Psi_q(\gamma, g) = \frac{1}{k} E \left[\left(\frac{g + kR_{n(k)}}{Z} - g \right)^q | R_{n(k)} = r_{n(k)} \right]$. For $q = 1, 2$ and $\varepsilon = 0$, the Bayes predictor of $R_{m(k)}$ reduces to

$$\begin{aligned} \Psi_1^*(\gamma, g) &= \frac{1}{n + \gamma - 1} \left((m + \gamma - 1)r_{m(k)} + (m - n)\frac{g}{k} \right), \\ \Psi_2^*(\gamma, g) &= \sqrt{\frac{1}{k} \left((g + kr_{m(k)})^2 \frac{(m + \gamma - 1)(m + \gamma - 2)}{(n + \gamma - 1)(n + \gamma - 2)} - 2g(g + kr_{m(k)}) \frac{m + \gamma - 1}{n + \gamma - 1} + g^2 \right)}. \end{aligned}$$

After some algebraic calculations, we observe that $\Psi_1^*(\gamma, g)$ and $\Psi_2^*(\gamma, g)$ are increasing in g and decreasing in γ . Hence, the PRGM predictors of $R_{m(k)}$ under the GEPLF (3) for $q = 1, 2$ and over the three classes of priors in Γ_g , Γ_{γ} and $\Gamma_{\gamma, g}$ are

$$R_{m(k)}^{PRGM, \Gamma} = \begin{cases} \left(\frac{\Psi_q^{*q}(\gamma_0, g_2) - \Psi_q^{*q}(\gamma_0, g_1)}{\ln \Psi_q^{*q}(\gamma_0, g_2) - \ln \Psi_q^{*q}(\gamma_0, g_1)} \right)^{\frac{1}{q}} & q = 1, 2, \Gamma = \Gamma_g, \\ \left(\frac{\Psi_q^{*q}(\gamma_1, g_0) - \Psi_q^{*q}(\gamma_2, g_0)}{\ln \Psi_q^{*q}(\gamma_1, g_0) - \ln \Psi_q^{*q}(\gamma_2, g_0)} \right)^{\frac{1}{q}} & q = 1, 2, \Gamma = \Gamma_{\gamma}, \\ \left(\frac{\Psi_q^{*q}(\gamma_1, g_2) - \Psi_q^{*q}(\gamma_2, g_1)}{\ln \Psi_q^{*q}(\gamma_1, g_2) - \ln \Psi_q^{*q}(\gamma_2, g_1)} \right)^{\frac{1}{q}} & q = 1, 2, \Gamma = \Gamma_{\gamma, g}. \end{cases}$$

Remark 1 Jafari Jozani and Parsian (2008) considered the GEPLF (3) for the special case of $q = 1$, i.e., the EPLF (2) and obtained the PRGM predictor of Y . It is interesting to mention that their prediction result can be obtained from our Example 2, when $q = 1$ and $\varepsilon = 1$, as an special case.

3. Application to Air Temperature Prediction Problem

In this section, we consider the air temperature prediction problem and apply the Bayes and robust Bayes prediction results in predicting some future k -records based on observed records from a Pareto model with pdf and survival function

$$f(x|\alpha, \eta) = \frac{\alpha \eta^\alpha}{x^{\alpha+1}}, \quad \bar{F}(x|\alpha, \eta) = \left(\frac{x}{\eta}\right)^\alpha, \quad x \geq \eta, \quad \alpha > 0, \quad \eta > 0. \quad (10)$$

The Pareto model has benefits in modeling some phenomena such as air temperatures, insurance claim sizes and stock price fluctuations, see Wilks (1959) and Madi and Raqab (2004).

Bayesian prediction of temperature records for the Pareto model was considered by Madi and Raqab (2004), but their study is under the symmetric squared error prediction loss and the obtained Bayes predictors are future 1-records. Here, we consider the performance of both Bayes and PRGM k -record predictors over some chosen classes of priors under the asymmetric GEPLF. Let $X_1, X_2, \dots$ be a sequence of *i.i.d* random variables from a Pareto distribution with unknown shape parameter α and unknown scale parameter η .

Following Arnold et al. (1998), after some manipulations, we observe that

$$\begin{aligned} f(r_{n(k)}|\alpha, \eta) &= \frac{(k\alpha)^n \eta^{k\alpha}}{\Gamma(n) r_{n(k)}^{k\alpha+1}} \left(\ln \frac{r_{n(k)}}{\eta}\right)^{n-1}, \\ f(r_{n(k)}, r_{m(k)}|\alpha, \eta) &= \frac{(k\alpha)^m \eta^{k\alpha}}{\Gamma(n) \Gamma(m-n) r_{n(k)}^{k\alpha+1} r_{m(k)}^{k\alpha+1}} \left(\ln \frac{r_{n(k)}}{\eta}\right)^{n-1} \left(\ln \frac{r_{m(k)}}{r_{n(k)}}\right)^{m-n-1} \end{aligned}$$

provided $r_{m(k)} > r_{n(k)} > \eta$. Now, in the ε -contaminated class of priors Γ_ε , let $\varepsilon = 0$ and $\pi_0 = \pi^\dagger$, where $\pi^\dagger$ is the generalized Lwin (1972) prior (also known as power-Gamma prior)

$$\pi^\dagger(\alpha, \eta) \propto \alpha^\nu \eta^{\lambda\alpha-1} \mu^{-\alpha}, \quad \alpha > 0, \quad 0 < \eta < \eta^*, \quad (11)$$

with positive constant numbers ν, λ, μ and η^* . Then

$$\pi^\dagger(\alpha, \eta|r_{n(k)}) \propto \frac{\alpha^{n+\nu}}{\eta} e^{-\alpha(\Lambda_1 - (\lambda+k)\ln \eta)}, \quad \eta < M, \quad (12)$$

where $M = \min\{r_{n(k)}, \eta^*\}$ and $\Lambda_1 = \ln \mu + k \ln r_{n(k)}$,

$$f(r_{m(k)}|r_{n(k)}, \alpha, \eta) = \frac{(k\alpha)^{m-n}}{\Gamma(n-m)} \frac{1}{r_{m(k)}} \left(\frac{r_{n(k)}}{r_{m(k)}}\right)^{k\alpha} \left(\ln \frac{r_{m(k)}}{r_{n(k)}}\right)^{m-n-1}. \quad (13)$$

Therefore, using (12) and (13), the predictive density (1) is obtained as below

$$f(r_{m(k)}|r_{n(k)}) = \frac{k^{m-n}}{r_{m(k)}} \frac{(\Lambda_2 + k \ln r_{n(k)})^{n+\nu}}{B(n+\nu, m-n)} \frac{(\ln r_{m(k)} - \ln r_{n(k)})^{m-n-1}}{(\Lambda_2 + k \ln r_{m(k)})^{m+\nu}}, \quad (14)$$

where $\Lambda_2 = \ln \mu - (\lambda + k) \ln M$ and $r_{m(k)} > r_{n(k)} > \eta$. To derive the Bayes predictors of $R_{m(k)}$, let $W = \ln \frac{R_{m(k)}}{R_{n(k)}}$, then the Bayes predictor of W under the GEPLF is

$$W^*(a, b, c, q, k) = (E[W^q | R_{n(k)} = r_{n(k)}])^{\frac{1}{q}} = \frac{b}{k} \left(\frac{\Gamma(c+q)\Gamma(a-q)}{\Gamma(c)\Gamma(a)} \right)^{\frac{1}{q}},$$

where $a = n + v$, $b = \Lambda_2 + k \ln r_{n(k)}$, $c = m - n$. Hence, the Bayes predictor of $R_{m(k)}$ is given by $R_{n(k)} e^{W^*}$.

To derive the PRGM predictors, in the ε -contaminated class of priors Γ_ε , let $\varepsilon = 1$, $\pi = \pi^\dagger$ and $\Gamma = \Gamma^\dagger$, where $\Gamma^\dagger$ is a given class of power-Gamma priors.

Now, to illustrate the prediction procedure, we use the average July temperatures (in degrees centigrade) of Neuenburg, Switzerland, during the period 1864-1993 (from Kluppelberg and Schwere, 1995). Table 1 summarizes 1-, 2- and 3-record values when a new record value was set.

Table 1: k -record temperature values.

Year	1864	1865	1866	1868	1869	1870	1873	1874	1881	1904	1911	1928	1952	1983
1-records	19.0	20.1	—	—	—	21.4	—	—	21.7	22.0	22.1	22.6	—	23.4
2-records	—	20.1	—	—	21.0	21.4	—	—	21.7	22.0	22.1	22.6	22.3	23.4
3-records	—	—	18.4	19.7	21.0	21.4	20.4	20.9	21.7	22.0	22.1	22.6	22.3	23.4

To specify the hyperparameter values, assume for example, the mean of α is 40 and its variance is 20, and the median and the 95th percentile of η are respectively 20 and 25. Then, after some manipulations, one can observe that $v = 80$, $\lambda = 0.072$, $\eta^* = 24.44$ and $\mu = 9.34$, see Madi and Raqab (2004). Now, for taking into account the prior uncertainty, assume the following classes of power-Gamma priors

$$\Gamma_1^\dagger = \left\{ \pi^\dagger(\alpha, \eta | v, \lambda, \mu, \eta^*) : v = 80, 0.06 \leq \lambda \leq 0.08, 8 \leq \mu \leq 10, \eta^* = 24.44 \right\}, \quad (15)$$

$$\Gamma_2^\dagger = \left\{ \pi^\dagger(\alpha, \eta | v, \lambda, \mu, \eta^*) : 70 \leq v \leq 90, 0.06 \leq \lambda \leq 0.08, 8 \leq \mu \leq 10, \eta^* = 24.44 \right\}. \quad (16)$$

Table 2 summarizes values of Bayes predictor ($R_{m(k)}^B$), PRGM predictor over $\Gamma_1^\dagger$ ($R_{m(k)}^{PR, \Gamma_1^\dagger}$) and PRGM predictor over $\Gamma_2^\dagger$ ($R_{m(k)}^{PR, \Gamma_2^\dagger}$) of $R_{m(k)}$ based on observing $R_{n(k)}$ with selected values of k , q , n and m . From this table, some interesting conclusions are carried out. For example, based on observing $R_{6(1)} = 22.1$, the future Bayes temperature value of $R_{10(1)}$ under GEPLF with $q = -1$ and $q = 1$ is predicted to be 23.71 and 24.29, respectively. This shows that the resulting prediction value when considering the GEPLF with $q = -1$ is less than the prediction value choosing $q = 1$. This relation holds for other prediction values in Table 2 too. The future Bayes temperature value of $R_{10(1)}$ is also computed based on observing $R_{8(1)} = 23.4$. It is seen that under GEPLF with $q = 1$, $R_{10(1)}$ is predicted to be 24.51 and this value is close the Bayes temperature value by $R_{6(1)}$. Similar conclusions are derived for PRGM temperature values. It is worth noting that $R_{8(k)}$, $k = 1, 2, 3$, is a real k -record, as reported in Table 2, and the predicted values in Table 2 based on observing $R_{6(k)}$ are quite close to its reported values. For example $R_{8(3)}$ is reported by 22 and based on observing $R_{6(3)} = 20.9$, the Bayes and PRGM temperature values under the GEPLF with $q = 1$ are $R_{8(3)}^B = 21.23$, $R_{8(3)}^{PR, \Gamma_1^\dagger} = 21.25$ and $R_{8(3)}^{PR, \Gamma_2^\dagger} = 21.25$. It is seen that for other values of k and m , the PRGM temperature prediction values based on observing $R_{6(k)}$, in comparison with the Bayes temperature prediction values, are closer to the real one reported in Table 1.

Table 2: Values of Bayes and PRGM predictors of $R_{m(k)}$ based on observing $R_{n(k)}$ with selected values of k, q, n and m .

	$q = -1, n = 6$			$q = 1, n = 6$			$q = 1, n = 8$		
	m			m			m		
	8	9	10	8	9	10	8	9	10
$R_{m(1)}^B$	22.62	23.16	23.71	23.17	23.72	24.29	23.95	24.51	25.08
$PR_{\Gamma_1^\dagger} R_{m(1)}$	22.63	23.17	23.72	23.18	23.74	24.32	23.95	24.52	25.14
$PR_{\Gamma_2^\dagger} R_{m(1)}$	22.64	23.20	23.76	23.20	23.78	24.36	24.98	24.57	25.18
$R_{m(2)}^B$	22.26	22.52	22.79	22.53	22.80	23.07	22.86	23.13	23.40
$PR_{\Gamma_1^\dagger} R_{m(2)}$	22.27	22.54	22.81	22.55	22.82	23.10	22.87	23.14	23.43
$PR_{\Gamma_2^\dagger} R_{m(2)}$	22.27	22.55	22.83	22.58	22.84	23.13	22.88	23.16	23.44
$R_{m(3)}^B$	21.06	21.23	21.40	21.23	21.40	21.57	22.17	22.34	22.52
$PR_{\Gamma_1^\dagger} R_{m(3)}$	21.07	21.24	21.41	21.25	21.42	21.60	22.18	22.35	22.53
$PR_{\Gamma_2^\dagger} R_{m(3)}$	21.07	21.25	21.42	21.25	21.43	21.61	22.18	22.36	22.54

3.1. Illustrative Simulation

In this subsection, we conduct an illustrative simulation to observe the behavior of Bayes and PRGM predictors obtained in the Pareto model. For this purpose, we generate some data from the Pareto model (10) with some chosen shape and scale parameters and then, considering the GEPLF (3) with $q = -1, 1$, we predict $R_{m(1)}$, $m = 3, 4, 5$ based on observing $R_{1(1)}$. For prior elicitation, consider the power-Gamma prior (11) with $v = 80$, $\lambda = 0.072$, $\eta^* = 24.44$ and $\mu = 9.34$ (as chosen in our air temperature prediction problem), and for taking the prior variation into account, assume the two classes of power-Gamma priors (15) and (16), and denote the corresponding Bayes and PRGM predictors by $d_m[1] = R_{m(1)}^B$, $d_m[2] = R_{m(1)}^{PR, \Gamma_1^\dagger}$ and $d_m[3] = R_{m(1)}^{PR, \Gamma_2^\dagger}$, $m = 3, 4, 5$.

We proceed the simulation study as follows:

1. Generate a random sample $x_1, \dots, x_l$, $l = 10^4$, from Pareto model (10) with $(\alpha, \eta) = (40, 20)$, $(60, 30)$.
2. Specify $r_{1(1)}$, $r_{3(1)}$, $r_{4(1)}$ and $r_{5(1)}$.
3. Compute the Bayes and PRGM predictors of $R_{m(1)}$, i.e., $d_m[1]$, $d_m[2]$ and $d_m[3]$, for $m = 3, 4, 5$.
4. Calculate prediction loss for $r_{m(1)}$, i.e., $L(r_{m(1)}, d_m[l]) = \left(\frac{r_{m(1)}}{d_m[l]}\right)^q - q \ln \frac{r_{m(1)}}{d_m[l]} - 1$, $m = 3, 4, 5$, $l = 1, 2, 3$, when $q = -1, 1$.

5. Repeat Steps 1-4 for $N = 10^3$ times and let $L_i(r_{m(1)}, d_m[l])$ denote prediction loss for i -th repetition ($i = 1, \dots, N$). Calculate Estimated Average Prediction Risk (EAPR) as a measure of comparison study

$$EAPR(d[l]) = \frac{1}{N} \sum_{i=1}^N L_i(r_{m(1)}, d_m[l]),$$

for $m = 3, 4, 5$ and the selected values of q in Step 4.

The results are summarized in Table 3 for a comparison purpose. From this table, we observe that when the data are generated from the Pareto model (10) with $(\alpha, \eta) = (40, 20)$, the Bayes predictors have smaller EAPR values than the PRGM predictors and hence, the Bayes predictors act better than the PRGM predictors. This nice property stems from our justifiably chosen hyperparameters of the power-Gamma prior (11) with $\nu = 80$, $\lambda = 0.072$, $\eta^* = 24.44$ and $\mu = 9.34$. But when $(\alpha, \eta) = (60, 30)$, our hyperparameters are not justifiably chosen and we observe that PRGM predictors outperform the Bayes predictors. This is due to the fact that robust rules are aimed at global prevention against inappropriate choices of priors or its hyperparameters, and when we do not have a justifiable prior information, PRGM predictors are expected to act better than the Bayes predictors. However, for justifiably chosen hyperparameters, Bayes predictors outperform the PRGM predictors, as somehow expected.

Table 3: EAPR of Bayes and PRGM predictors for selected values of q , η and α .

$(\alpha, \eta) = (40, 20)$						
$q = -1$			$q = 1$			
m			m			
3	4	5	3	4	5	
$d_m[1]$	0.001000	0.001795	0.002075	0.000677	0.000879	0.000837
$d_m[2]$	0.001124	0.002165	0.002628	0.000697	0.001182	0.001375
$d_m[3]$	0.001106	0.002103	0.002518	0.000689	0.001125	0.001273
$(\alpha, \eta) = (60, 30)$						
$q = -1$			$q = 1$			
m			m			
3	4	5	3	4	5	
$d_m[1]$	0.000327	0.000279	0.000816	0.000381	0.000638	0.001490
$d_m[2]$	0.000302	0.000265	0.000743	0.000248	0.000340	0.000966
$d_m[3]$	0.000316	0.000268	0.000752	0.000263	0.000374	0.001026

4. Concluding Remarks

The following concluding remarks are noteworthy:

- (i) The Gamma model considered in Example 1 can be replaced by a general scale model and all the consequent predictors can be expressed in a general setting. To see this, let $X_1, X_2, \dots$ be a sequence of independent random variables from $\frac{1}{\tau}f(\frac{x}{\tau})$, with f is known and τ is an unknown scale parameter. In many cases the joint density reduces to

$$c(x, n) \eta^{-\alpha} e^{-T(x)/\eta},$$

where $c(x, n)$ is a function of $x = (x_1, \dots, x_n)$ and n , $\eta = \tau^r$ for some r , α is a function of n and $T(X)$ is a complete sufficient statistic for η with $\Gamma(\alpha, \eta)$ -distribution, see Parsian and Nematollahi (1996). Now, if $(X_{n+1}, \dots, X_{n+m})$ be an independent sequence of the future observations, then $X = T(X_1, \dots, X_n) \sim \Gamma(\alpha_1, \eta)$ and $Y = T(X_{n+1}, \dots, X_{n+m}) \sim \Gamma(\alpha_2, \eta)$ are independent. This fact provides an extension of our prediction results to a general class of scale model $\frac{1}{\tau}f(\frac{x}{\tau})$ populations.

- (ii) It is interesting to mention that in the context of robust Bayesian analysis, the sensitivity of choosing some predictors is measured by a recently introduced criteria by Arias-Nicolas, Martin, Ruggeri and Suarez-Llorens (2009) which in fact extends the criteria considered by Ruggeri and Sivaganesan (2000). When $r_p^*(d, d_\pi) = \frac{\rho(\pi, d) - \rho(\pi, d_\pi)}{\rho(\pi, d_\pi)}$, as sensitivity of choosing the predictor d instead of the Bayes predictor d_π , the predictor $d_{LS, \Gamma}$ is said to be a Least Sensitive (LS) predictor if $r_p^*(d_{LS, \Gamma}, d_\pi) = \inf_{d \in \mathcal{D}} \sup_{\pi \in \Gamma} r_p^*(d, d_\pi)$. When $\rho(\pi, d_\pi)$ does not depend on the hyperparameters in a chosen class Γ of priors, which occurs in Example 1, leads us to the fact that LS predictor is identical to the corresponding PRGM predictor. To see this, notice that the posterior function (4) is computed as

$$\begin{aligned} \rho(\pi, d_\pi) &= q \ln d_\pi - qE[\ln Y|X] \\ &= q \ln d_\pi - qE[[\ln Y|\eta]|X] \\ &= q \ln d_\pi - qE[\ln \eta|X] - q\Psi(\alpha_2) \\ &= q \ln \frac{d_\pi}{X+g} + q\Psi(\alpha_1 + \gamma) - q\Psi(\alpha_2), \\ &= q \ln \Psi(\alpha_1 + \gamma, \alpha_2, q) + q\Psi(\alpha_1 + \gamma) - q\Psi(\alpha_2), \end{aligned} \quad (17)$$

where $\Psi(\alpha_1 + \gamma) = \frac{d}{d\alpha_1} \ln \Gamma(\alpha_1 + \gamma)$ and $\Psi(\alpha_2) = \frac{d}{d\alpha_2} \ln \Gamma(\alpha_2)$. Obviously, the posterior function $\rho(\pi, d_\pi)$ in (17) does not depend on the values of g . This guarantees that $d_{LS, \Gamma_g} = d_{PRGM, \Gamma_g}$.

- (iii) It is noticeable that the Bayes predictor under squared error prediction loss function is identical to that under the GEPLF with $q = 1$ and hence, one might be interested in comparing our Bayesian prediction results with those obtained in Madi and Raqab (2004) for the case $k = 1$. This is possible but a correction in their results is needed. Their predictive density, given by

$$f(r_{m(1)}|r_{n(1)}) = \frac{1}{r_{m(1)}} \frac{(\Lambda_2 + \ln r_{m(1)})^{n+v}}{B(n+v, m-n)} \frac{(\ln r_{m(1)} - \ln r_{n(1)})^{m-n-1}}{(\Lambda_2 + \ln r_{n(1)})^{m+v}}, \quad (18)$$

where $\Lambda_2 = \ln \mu - (\lambda + 1) \ln M_2$, $M_2 = \min \{r_{n(1)}, M_1\}$, $M_1 = \min \{r_{1(1)}, \eta^*\}$, depends on $r_{1(1)}$ and this violates the Markovian property in k -record data. Besides, if $r_{1(1)}$ is an outlier, it might cause a serious problem in the prediction procedure. However, the predictive density (14), in comparison with the predictive density (18), is robust w.r.t. the effects of outliers, and adjusts this major flaw. This defect affects the prediction values and one should be cautious when comparing the prediction values obtained here with those reported there.

Appendix (Proof of Theorem 1)

Note that $\frac{\partial}{\partial d_{\pi_\epsilon}} r_p(d, d_{\pi_\epsilon}) = \frac{q}{d_{\pi_\epsilon}} \left(\frac{d_{\pi_\epsilon}^q}{d^q} - 1 \right)$ and $\frac{\partial^2}{\partial d_{\pi_\epsilon}^2} r_p(d, d_{\pi_\epsilon}) = \frac{q}{d_{\pi_\epsilon}^2} \left((q-1) \frac{d_{\pi_\epsilon}^q}{d^q} + 1 \right)$. So, $r_p(d, d_{\pi_\epsilon})$ has a unique minimum at $d_{\pi_\epsilon} = d$. Each of the following three steps leads us to the PRGM predictor of Y under the GEPLF (3). Notice that $\underline{d} < d_{\pi_\epsilon} < \bar{d}$.

Step 1: When $d < \underline{d}$, differentiating $r_p(d, d_{\pi_\epsilon})$ w.r.t. d_{π_ϵ} , it is easy to verify that

$$\sup_{\pi_\epsilon \in \Gamma_\epsilon} r_p(d, d_{\pi_\epsilon}) = r_p(d, \bar{d}).$$

Let $f_1(d) = r_p(d, \bar{d})$, then $f_1'(d) = -\frac{q}{d} \left(\frac{\bar{d}^q}{d^q} - 1 \right)$. So for all $d < \underline{d}$, $f_1(d)$ is decreasing in d and $\inf_{d < \underline{d}} f_1(d) = f_1(\underline{d})$. It means that

$$\inf_{d < \underline{d}} \sup_{\pi_\epsilon \in \Gamma_\epsilon} r_p(d, d_{\pi_\epsilon}) = r_p(\underline{d}, \bar{d}), \quad f_1(\bar{d}) = 0.$$

Step 2: When $d > \bar{d}$, differentiating $r_p(d, d_{\pi_\epsilon})$ w.r.t. d_{π_ϵ} , it is easy to verify that

$$\sup_{\pi_\epsilon \in \Gamma_\epsilon} r_p(d, d_{\pi_\epsilon}) = r_p(d, \underline{d}).$$

Let $f_2(d) = r_p(d, \underline{d})$, then $f_2'(d) = -\frac{q}{d} \left(\frac{\underline{d}^q}{d^q} - 1 \right)$. So for all $d > \bar{d}$, $f_2(d)$ is increasing in d and $\inf_{d > \bar{d}} f_2(d) = f_2(\bar{d})$. It means that

$$\inf_{d > \bar{d}} \sup_{\pi_\epsilon \in \Gamma_\epsilon} r_p(d, d_{\pi_\epsilon}) = r_p(\bar{d}, \underline{d}), \quad f_2(\underline{d}) = 0.$$

Step 3: When $\underline{d} < d < \bar{d}$, there are two possibilities: either $\underline{d} < d < d_{\pi_\epsilon} < \bar{d}$ or $\underline{d} < d_{\pi_\epsilon} < d < \bar{d}$. So for these possibilities, $\sup_{\pi_\epsilon \in \Gamma_\epsilon} r_p(d, d_{\pi_\epsilon}) = \max \{r_p(d, \bar{d}), r_p(d, \underline{d})\}$. Let

$$l(d) = f_1(d) - f_2(d) = \frac{\bar{d}^q - \underline{d}^q}{d^q} - q \ln \frac{\bar{d}}{d}.$$

It is easy to verify that $l(d)$ is continuous and decreasing in d and $l(\underline{d})l(\bar{d}) < 0$. Thus, there exists $d^* \in (\underline{d}, \bar{d})$ such that $l(d^*) = 0$, i.e., $r_p(d^*, \underline{d}) = r_p(d^*, \bar{d})$. Therefore for all $\underline{d} < d < d^*$, $l(d) > 0$ and $\max \{r_p(d, \bar{d}), r_p(d, \underline{d})\} = r_p(d, \bar{d})$. Furthermore for all $\underline{d} < d < d^*$, $r_p(d, \bar{d})$ is decreasing in d and

$$\inf_{\underline{d} < d < d^*} r_p(d, \bar{d}) = r_p(d^*, \bar{d}).$$

On the other hand, for all $d^* < d < \bar{d}$, $l(d) < 0$ and $\max \{r_p(d, \bar{d}), r_p(d, \underline{d})\} = r_p(d, \underline{d})$. Hence for all $d^* < d < \bar{d}$, $r_p(d, \underline{d})$ is increasing in d and

$$\inf_{d^* < d < \bar{d}} r_p(d, \underline{d}) = r_p(d^*, \underline{d}).$$

Combining these arguments leads us to

$$\inf_{\underline{d} < d < \bar{d}} \sup_{\pi_e \in \Gamma_e} r_p(d, d_{\pi_e}) = r_p(d^*, \underline{d}) = r_p(d^*, \bar{d}).$$

Using the above three steps, it is easy to see that

$$\inf_{d \in \mathcal{D}} \sup_{\pi_e \in \Gamma_e} r_p(d, d_{\pi_e}) = \inf_{\underline{d} < d < \bar{d}} \sup_{\pi_e \in \Gamma_e} r_p(d, d_{\pi_e}) = r_p(d^*, \bar{d}) = r_p(d^*, \underline{d}).$$

Now, for $d_{PRGM, \Gamma_e}(X) = d^* \in (\underline{d}, \bar{d})$, $l(d_{PRGM}(X)) = 0$.

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